



Graphing Network Performance with MRTG

This article guides readers on how to use MRTG in order to display the ADSL traffic of a Cisco 877W router with a little help from SNMP (Simple Network Management Protocol).

MRTG (Multi Router Traffic Grapher), a tool for graphing numerical data, is free and has been written in Perl by Tobi Oetiker. Let's first check out how much fun installing MRTG can be.

Your Linux distribution will probably include an MRTG package so you will not have any problems installing it. At the time of writing this article, the latest MRTG stable release is version 2.17.4.

Configuring SNMP on a Cisco 877W ADSL router

The first thing to do on the Cisco 877W ADSL router is to turn on SNMP. Then create a SNMP community string that will help you get the required information.

The following commands should be executed on the Cisco router using IOS:

```
cisco877w#enable
cisco877w#show snmp
%SNMP agent not enabled
cisco877w#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
cisco877w(config)#snmp-server community OpenSourceFY RO
cisco877w(config)#exit

cisco877w#write memory
Building configuration...
[OK]
cisco877w#
```

The RO community string (OpenSourceFY) permits *Get* requests only, whereas a RW community string (that is required for this article) allows both *Get* and *Set* requests. Allowing *Set* requests without any reason and without using Cisco Access Lists (ACL) is a security threat.

The following command, that produces plenty of output, will examine if the SNMP router set-up is working as expected:

```
$ snmpwalk -Os -c OpenSourceFY -v 1 cisco
```

The *cfgmaker* utility

The *cfgmaker* MRTG utility helps create configuration files. It may add unnecessary data that you should clean up later, but its output is a great starting point.

I ran the *cfgmaker* utility as follows:

```
$ cfgmaker --global 'WorkDir: /Users/mtsouk/Sites/mrtg'
OpenSourceFY@cisco > cisco-MRTG.cfg
```

The executed command creates the *cisco-MRTG.cfg* file inside the */Users/mtsouk/Sites/mrtg* directory and queries a machine named 'cisco' (I have a 'cisco' entry in my */etc/hosts* file) using the 'OpenSourceFY' community string.

After creating the configuration file, a Web page that will allow access to MRTG's output should be created. This task can be done using the *indexmaker* utility, which also comes with MRTG, as follows: